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Journal:	<i>Applied Physiology, Nutrition, and Metabolism</i>
Manuscript ID	apnm-2024-0191.R2
Manuscript Type:	Research Article
Date Submitted by the Author:	02-Sep-2024
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Is the manuscript for consideration in a Special Issue or Collection?:	The Physiology of Extreme Temperatures
Keyword:	aging, heat stress, climate change, heart rate variability, cardiac autonomic modulation, autonomic nervous system < nervous system

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**An exploratory investigation of heart rate and heart rate variability responses to
daylong heat exposure in young and older adults**

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ABSTRACT

Heart rate variability (HRV) has shown potential as a tool for monitoring thermal strain, but there is limited data to support its efficacy in older adults during prolonged heat exposures. We compared HRV between young (19-31 years, n=20) and older (61-78 years, n=39) adults during 9 hours of heat exposure (40°C, 9% RH). We also explored whether heart rate (HR) and/or HRV could be used to distinguish older adults who achieved elevated thermal strain, defined as either 1) an increase in core temperature $>1.0^{\circ}\text{C}$ (occurring in 39% [15/39]) or 2) a reduction in systolic blood pressure >10 mm Hg (occurring in 67% [26/39]). Percentage of age-predicted maximal HR and percentage of heart rate reserve (HRR) were higher, whereas standard deviation of normal RR intervals (SDNN), the square root of the mean of squared differences between successive RR intervals (RMSSD), high frequency power (HF), and cardiac vagal index (CVI) were lower in older compared to young adults ($P\leq 0.004$) during heat exposure. In older adults, increases in core temperature were correlated with percentage of age-predicted maximal HR, percentage of HRR, RMSSD, and CVI ($P\leq 0.031$), whereas changes in systolic blood pressure were not significantly associated with HR or HRV indices ($P\leq 0.327$). Receiver operating characteristic curve analysis indicated that HR and HRV indices had generally poor ability to identify older adults with elevated thermal strain (area under the curve ≤ 0.65). Age-related differences in HRV, consistent with vagal withdrawal among older adults, remained during daylong heat exposure, but marked heterogeneity of response likely contributed to HRV providing limited discriminatory value in identifying changes in core temperature or blood pressure in older adults.

KEYWORDS: aging, heat stress, climate change, heart rate variability, cardiac autonomic modulation, autonomic nervous system

41 **Take-Home Message**

42 Age differences in heart rate variability (HRV), reflecting vagal withdrawal in older versus young
43 adults, were stable during daylong heat exposure, while the utility of HRV as a tool to identify
44 older adults with elevated thermal strain was limited.

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INTRODUCTION

Older individuals are among the most vulnerable to heat-related illness and hospitalization due, in large part, to greater rises in body temperature during heat exposure secondary to blunted increases in sweat secretion, skin blood flow, and cardiac output relative to their younger counterparts (Meade et al. 2020; Arsad et al. 2022). Since lethal heat is expected to occur with increasing regularity (Guo et al. 2018), developing evidence-based, non-invasive heat strain monitoring strategies to facilitate early heat-illness detection and intervention is paramount (Kuras et al. 2017; Pham et al. 2020).

Body temperature regulation depends on efficient functioning of thermoregulatory processes coordinated by the autonomic nervous system (Ferreira et al. 2023). Liu and colleagues measured heart rate variability (HRV), a non-invasive indicator of cardiac autonomic heart rate modulation, in college students during a short exposure to varying ambient temperatures, and observed a higher ratio of absolute power in low- and high-frequency bands (i.e., LF/HF) at elevated levels of thermal discomfort as a function of increasing ambient temperatures (21-30°C) (Liu et al. 2008). Based on these findings, the authors suggested that HRV may provide non-invasive insight into body temperature regulation. In another study, De Barros et al. (2024) provided preliminary evidence supporting the utility of heart rate and, to a lesser extent, HRV metrics such as the square root of the mean of squared differences between successive RR intervals (RMSSD) and detrended fluctuation analysis (DFA) alpha 1 as potential predictors of heat tolerance. While the aforementioned studies support the potential utility of HRV for non-invasive physiological heat strain monitoring, they were performed in young adults while resting (Liu et al. 2008) or in individuals performing simulated work bouts in the heat (De Barros et al. 2024). There is some evidence, although indirect, to suggest that HRV may be useful for detecting age-related

increases in heat strain under conditions more reflective of those experienced during heat waves (Carrillo et al. 2016, 2019; Meade et al. 2023a). For example, we recently observed that marked reductions in core temperature induced by 2 hours of air-conditioning midway through a 9-hour simulated heat wave (40°C, 9% relative humidity (RH)) were paralleled by reductions in heart rate and improved RMSSD in older adults (~72 years) (Meade et al. 2023a). However, as shown in our recent work, age-related alterations in HRV during passive heat stress have not been a universal finding (Carrillo et al. 2016, 2019). For example, we reported that, despite elevated levels of thermal strain among older adults (Stapleton et al. 2014), age-related reductions in HRV did not worsen during brief 120-minute passive exposures to humid (36.5°C, 60% RH) and dry (36.5°C, 20% RH) heat (Carrillo et al. 2019).

There is currently limited data on age-related differences in cardiac autonomic heart rate modulation assessed during prolonged heat exposures reflective of those experienced during heat waves. It is also unclear whether HRV monitoring is an efficacious strategy for determining thermal strain in older adults. We therefore compared HRV between older and young adults during a daylong (9 hours) heat exposure similar to indoor conditions reported during recent heat events in North America (40°C, 9% RH) (Henderson et al. 2022). We also explored whether HRV metrics could be used to distinguish different levels of physiological heat strain (i.e., core temperature and blood pressure responses) in older adults. This exploratory study informs the need for larger investigations on age-related changes in HRV during prolonged heat exposure and should be considered hypothesis generating. As such, no formal hypotheses were tested.

MATERIALS AND METHODS

This study is part of a larger project that investigated the physiological responses to daylong heat exposure in young and older adults (ClinicalTrials.gov identifier: NCT04353076) (Meade et al. 2023b). This study was approved by the University of Ottawa Health Sciences and Science Research Ethics Board (H-11-18-1186) and was conducted according to the standards set by the Declaration of Helsinki. Aspects of the methodology pertinent to the current analysis are provided below; readers are referred to the primary report for full details (Meade et al. 2023b).

Twenty young (24 (4) years) and 39 older adults (71 (4) years) with (n=20) and without (n=19) type 2 diabetes and/or hypertension provided a written statement of informed consent prior to participation in this study. Individuals were excluded if they were experiencing physical restriction that would limit successful implementation of the study protocol, or if using medication that would be inadvisable based on study procedures (see Meade et al. 2023b for more details regarding reported prescription medications). Participant characteristics are provided in Table 1. Participants were asked to avoid physical activity and alcohol consumption during the 24 hours prior to experimental testing. To ensure adequate hydration, participants were asked to consume a minimum of 500 mL of water on the night before and the morning of the experimental session. Participants were also asked to consume a light meal ~2 hours prior to arriving at the laboratory for testing. On the day of the experimental session, participants arrived at the laboratory between 06:30-09:00. They were instructed to insert a general-purpose thermocouple temperature probe ~12 cm past the anal sphincter for measurement of rectal temperature (Mon-a-therm General Purpose Temperature Probe, Mallinckrodt Medical Inc.), and then were fitted with a Holter Monitor (DigiTrak XT Holter Monitor, Philips) for continuous HRV assessment. Following a 2-hour period for instrumentation and baseline assessments, which included 10 min of quiet rest for

stable HRV measurement (spontaneous breathing), participants entered a climate-controlled chamber wearing sandals, shorts, and a light top for women, to begin the 9-hour heat exposure (40°C, 9% RH; 25.5°C wet-bulb globe temperature (WBGT)). Ten minutes of resting HRV measurements were completed every 3 hours during heat exposure (i.e., at hours 3, 6, and 9). Participants remained seated for the majority of the heat exposure period. During hours 4-6, a light, self-provided lunch (e.g., peanut butter sandwich) was consumed and water was available *ad libitum*. During this time, participants were permitted to stand and perform light stretching if desired.

Heart rate was extracted, and RR intervals were derived from raw data (3 channels at a sampling rate of 175 Hz) recorded using the Holter monitor. Data were analyzed using the Philips Zymed Software (Philips Zymed Version 3.0, Andover, MA), and only beats considered normal by the Zymed algorithm were used for further analysis. Continuous Individualized Variability Analysis (CIMVA) software was used to extract five HRV indices using the normal-to-normal interval time series and a 5 min windowed analysis with 30 s time step (Herry et al. 2013). Time-domain indices included the standard deviation of normal RR intervals (SDNN) and the square root of the mean of squared differences between successive RR intervals (RMSSD) – two indices primarily influenced by the parasympathetic nervous system (Michael et al. 2017). Frequency-domain measurement included the high (HF) (0.15 – 0.40 Hz) frequency band (in ms^2) – a marker primarily influenced by vagal activity (Malik et al. 1996). Detrended fluctuation analysis (DFA) was used to calculate short-term ($\alpha 1$) fractal relationship properties of RR intervals (Peng et al. 1995). This clinically relevant metric is sensitive to pathological processes and has been described as an objective, potential predictor of mortality (Peng et al. 1995; Andrade et al. 2023). Cardiac vagal index was calculated ($\text{CVI} = \log_{10} [16 * \text{SD1} * \text{SD2}]$) as a non-linear method measured to

assess cardiac autonomic function that is sensitive to vagal modulation (Toichi et al. 1997; Macartney et al. 2020). Changes in heart rate were evaluated in absolute values, as a percentage of age-predicted maximum ($220 - \text{age}$), and as a percentage of heart rate reserve (age-predicted maximum heart rate - resting heart rate). Heart rate and heart rate variability data were smoothed (Meade et al. 2023a) and extracted as minute data, and at each of the four resting measurement periods occurring at ~3-hour intervals.

Rectal temperature data were recorded using LabVIEW software (Version 7.0, National Instruments) in 15 s intervals. Due to technical difficulties or participant refusal to insert the rectal temperature probe, core temperature measurement in 4 participants was estimated using a temperature capsule (VitalSense® ingestible capsule thermometer; Mini Mitter Company) ingested upon arrival at the laboratory. Core temperature data were manually cleaned, interpolated, and converted to 15-minute averages at the end of each hour during the heat exposure period. Arterial systolic and diastolic blood pressures were taken as the average of three measurements via manual auscultation of the brachial artery (30 s between measurements).

Statistical Analysis

Heart rate and HRV indices were compared between the young and older adults using linear mixed-effects models both unadjusted and adjusted for baseline values. In the unadjusted models, fixed effects included age group (young and older), time (rates: hour 0, 3, 6, 9) and their interaction. In the adjusted models, baseline time point was instead modeled as a continuous covariate. Descriptive summaries of raw data are presented as means (standard deviations) and estimated between-group differences as means [95% confidence limits]. $P < 0.050$ (two-sided) was considered

statistically significant. Data were analyzed using R (Version 4.2.0, R Core Team) (Arel-Bundock 2022; Bates et al. 2010).

To evaluate the potential utility of heart rate and variability indices for quantifying the level of thermal strain in older adults, we first evaluated the slope of the relation between each heart rate and HRV index (continuous explanatory variable) and changes in core temperature and systolic blood pressure from basal values (continuous outcome variable, measured at 3-, 6-, and 9-hours during exposure) using linear mixed-effects models. Each model also included a random intercept to account for the multiple data points provided by each participant. We also constructed receiver operating characteristics (ROC) curves for each model to evaluate the utility of heart rate (absolute values, percent maximum, percent reserve) and HRV indices (SDNN, RMSSD, CVI, HF power, and DFA α_1) measured over the final 3 hours of exposure for identifying elevated levels of heat-induced physiological strain (i.e., heat-vulnerable older adults). Elevated thermal strain during heat exposure was characterized as an increase in core temperature of $>1.0^{\circ}\text{C}$ or a reduction in systolic pressure of >10 mm Hg (WHO Scientific Group 1969). The area under ROC curve (AUC) and its 95% confidence interval were used to quantify the ability of heart rate and HRV indices to discriminate between low/moderate and elevated physiological strain in older adults exposed to the heat. Note that because this was an exploratory investigation, no formal power calculation was performed. However, the produced confidence intervals provide readers a means to interpret the uncertainty in our estimates of AUC. We assessed the utility of each index in reference to normative ranges – in general an AUC of 0.5 suggests no discriminatory value, 0.7 to 0.8 is considered acceptable, 0.8 to 0.9 is considered excellent, and more than 0.9 is considered outstanding (Hosmer and Lemeshow 2000). The R packages *pROC* and *plotROC* were used for analysis and data visualization (Robin et al. 2011; Sachs 2017).

RESULTS

Pairwise comparisons for heart rate and HRV indices between young and older adults are shown in Table 2. Absolute heart rate was not different between young and older adults in either unadjusted or baseline-adjusted analyses (effect of age: $P \geq 0.311$). When expressed as a percentage of age-predicted max (age by time interaction: $P = 0.011$), however, heart rate was elevated in older adults at all time points (all $P < 0.001$), though these between-condition differences were no longer statistically significant after adjustment for baseline values (effect of age: $P = 0.292$). Consistent with the unadjusted analysis of maximal heart rate, older participants' heart rate was elevated an average of 8% heart rate reserve (effect of age: $P = 0.006$). In unadjusted analyses, log-transformed SDNN, log-transformed RMSSD, log-transformed HF power, and CVI were lower, on average, 0.6 [0.4, 0.8], 0.6 [0.3, 0.9], 0.4 [0.1, 0.6] and 0.5 [0.4, 0.7], respectively, in older compared to young adults (effect of age: all $P \leq 0.004$; note: P-values for age by time interactions for SDNN, RMSSD and CVI ranged from 0.054-0.070). However, age group differences for each variable were no longer statistically significant after adjustment for baseline values (effect of age: all $P \geq 0.129$). Finally, DFA α_1 (age by time interaction: $P = 0.002$) was not different between age groups at any time point in unadjusted analysis (all $P \geq 0.06$), but was significantly lower by 0.1 [0.1, 0.2] in older compared to young adults during heat exposure after adjustment for baseline values (effect of age: $P = 0.003$).

Associations between heart rate/HRV metrics and core temperature in older adults are shown in Figure 1. The change in core temperature from baseline values was significantly associated with percentage of age-predicted maximum heart rate, percentage of heart rate reserve,

log RMSSD, and CVI ($P \leq 0.031$). No indices of heart rate or HRV were significantly related to the change in systolic blood pressure ($P \geq 0.327$).

Figure 2 displays ROC curves for heart rate/HRV metrics conducted to further assess their utility in discriminating between different levels of core temperature and blood pressure responses during heat exposure in older participants. For core temperature (Figure 2A), AUC values ranged from 0.49-0.57, indicating a limited discriminatory value of heart rate/HRV metrics in differentiating the 39% (15/39) of older participants who experienced a core temperature increase of $>1^{\circ}\text{C}$. Similarly, AUC values calculated for discriminating between differences in systolic blood pressure ranged from 0.49-0.65 revealing a similar outcome as core temperature (Figure 2B) – a limited discriminatory value of heart rate/HRV metrics in differentiating the 67% (26/39) of older participants who experienced a systolic blood pressure decrease of >10 mm Hg during heat exposure.

DISCUSSION

We investigated HRV in young and older adults during daylong (9 hours) heat exposure that was similar to indoor ambient conditions experienced during recent heat events in North America (40°C , 9% RH) (Henderson et al. 2022). We found that older adults had a higher relative heart rate ($\sim 10\%$ when expressed as a percentage of age-predicted maximum) and lower levels of HRV, including SDNN, RMSSD, HF, and CVI throughout the experimental period (i.e., from baseline to end of heat exposure). These HRV metrics primarily reflect cardiac vagal activity and are consistently shown to be lower in older compared to young adults during resting, thermoneutral conditions (Almeida-Santos et al. 2016). During passive heat exposure, however, the limited evidence of age-related differences in cardiac autonomic heart rate modulation show some

variability (Carrillo et al. 2016, 2019). In the current study, when heart rate and HRV metrics were adjusted for baseline values, the age-related differences in heart rate, SDNN, RMSSD, HF, and CVI during heat exposure were no longer statistically significant, suggesting that both age groups experienced HRV changes associated with vagal withdrawal and sympathetic activation. The exception was DFA $\alpha 1$, which was statistically significantly different between age groups when adjusted for baseline levels.

A DFA $\alpha 1$ value of 1 reflects fractal-like behavior or more complex heart rhythms, whereas values that move further away from 1 in either direction reflect less complexity of heart rhythms or lower overall HRV (Heffernan et al. 2008; De Barros et al. 2024). In response to the daylong heat exposure, young participants' DFA $\alpha 1$ increased by an average of 2.7 units compared to baseline, whereas older participants increased by an average of 1.2 units. These results suggest greater vagal withdrawal and increased sympathetic modulation among younger participants during heat exposure when compared to older adults. These results are consistent with our prior findings (Carrillo et al. 2016), and support previous work indicating a heat-induced thermoregulatory response that is associated with increased vagal withdrawal and sympathetic dominance in young healthy adults (Crandall and Wilson 2014; Abellán-Aynés et al. 2021) that has been linked to the initiation of heat loss responses (Low et al. 2011). Thus, the smaller change in DFA $\alpha 1$ among older participants in response to the heat may reflect a blunted thermoregulatory response consistent with greater increases in body heat storage and core temperature when compared to younger participants as reported previously (Meade et al. 2023b).

We also assessed the utility of heart rate and HRV metrics for the discrimination of heat-induced physiological strain (i.e., thermal strain defined as an increase of $>1^{\circ}\text{C}$ in core temperature or a decrease of >10 mm Hg in systolic blood pressure) in heat-vulnerable older adults. The

percentage of older adults in our sample achieving an increase in rectal temperature $>1.0^{\circ}\text{C}$ and reduction in systolic blood pressure >10 mm Hg was 39% and 67%, respectively. Analyses revealed a significant association between core temperature rise and heart rate (percent of age-predicted maximum and reserve), RMSSD, and CVI, whereas no heart rate or HRV indices were related to a decrease in systolic blood pressure. AUC values calculated from ROC curves did not exceed 0.65, with lower and upper confidence limits from ≥ 0.3 to ≤ 0.8 , respectively. Despite the observed associations between changes in HRV and increases in core temperature among older participants, low AUC values suggest limited discriminatory utility of heart rate and HRV metrics in identifying older individuals experiencing elevated physiological strain during heat exposure, though, given the wide confidence intervals, more data is required to confirm this.

This study has limitations. Despite the prolonged (i.e., 9 hours) heat exposure duration implemented in the current study, our findings were limited to HRV responses observed during a single day of heat exposure. There is a growing concern about the cumulative heat-related deficits occurring over multiple days of heat stress (Notley et al. 2018a, 2018b). Thus, this work should be extended over several days to assess the cumulative impact of heat exposure on HRV and its relation to heat-induced physiological strain in older adults. Further, given that relative humidity in the present study was 9%, our findings may not be generalizable to conditions of extreme heat with high humidity. Confirmatory studies should incorporate higher levels of humid heat, in consideration of age-related differences in HRV responses (Carrillo et al. 2019), and the extensive physiological strain experienced by older adults during brief humid heat exposures (Stapleton et al. 2014).

While marked heterogeneity of response likely contributed to HRV providing limited discriminatory value in identifying elevated thermal strain among older adults, this would likely

be consistent with responses observed during extreme heat events experienced by heat-vulnerable older adults worldwide. Thus, a discriminatory tool aiming to identify older adults experiencing heat-induced physiological strain should account for heterogeneity of HRV responses. While by design, our study sample of older adults included males and females between the ages of 61 and 78 years, with or without common chronic health conditions (i.e., type 2 diabetes or hypertension), additional studies assessing the separate effects of these and other chronic diseases in men and women are necessary. The data generated from these studies could contribute to further understanding the utility of HRV in discriminating levels of physiological strain during prolonged heat exposures. This is particularly relevant since older adults with chronic disease, when compared to older adults without chronic disease, as well as older women, may experience altered cardiovascular responses to prolonged heat exposure which is in part mediated by sex- and disease-associated reductions in heat loss capacity (Meade et al. 2024; Hart et al. 2009). This would include assessing the impact of physical activity (such as activities of daily living) which may influence these responses (Grässler et al. 2021).

Time lags between heat-induced changes in HR/HRV and in core temperature were not considered. It is expected, however, that this did not significantly influence the results since HR/HRV and core temperature were averaged over a 10- and 15-minute time periods, respectively, at baseline and every 3 hours during the daylong heat exposure. Our study did not incorporate an exercise test to determine maximum heart rate. Instead, to estimate maximum heart rate, we used the standard equation $220 - \text{age in years}$, which provides only a rough estimate (Tanaka et al. 2001) and does not account for a potentially lower value among those with chronic disease (Lauer et al. 1996). However, an important benefit of using age-predicted max heart rate is its simplicity. This calculation, and others like the Tanaka et al. (2001) formula (predicted maximum heart rate = 208

296 - 0.7 x age), could easily be incorporated into wearable technology or other physiological monitors.
297 Determining which method of maximal heart rate estimation is more closely related to changes in
298 heat strain is an important avenue for future research. Finally, the small numbers of this study do
299 not allow for a definitive predictive modelling evaluation of the value of HRV in identifying the
300 thermal strain.

301 To our knowledge, these are the first data collected examining heart rate and HRV
302 responses in older adults exposed to daylong heat with the aim of examining their utility for
303 distinguishing elevated heat-induced physiological strain in this heat-vulnerable population. Our
304 findings reveal changes in HRV consistent with vagal withdrawal during daylong heat exposure
305 among older participants compared to their younger counterparts. These age-related HRV
306 differences during heat exposure may reflect blunted thermoregulatory responses in older adults
307 that could contribute to greater heat storage and a higher risk of heat-related illness. Our
308 exploratory study revealed that the utility of heart rate and HRV metrics, however, was limited in
309 being able to identify older adults experiencing elevated heat-induced physiological strain during
310 the daylong heat exposure.

311 **COMPETING INTERESTS:** The authors declare there are no competing interests.

312

313 **AUTHORSHIP CONTRIBUTION:** Conceptualization, RDM and GPK; Data Curation and
314 Formal Analysis, RDM, CLH, and GPK; Writing – Original Draft, AEC, RDM, and GPK; Writing
315 – Review & Editing, AEC, RDM, CLH, AJES, and GPK; Supervision, GPK; Project
316 Administration, Funding Acquisition, GPK.

317

318 **FUNDING:** This research was funded by the Canadian Institutes for Health Research (Grant No.
319 399434) and Health Canada (Contract No. 4500387992; to GPK).

320

321 **DATA AVAILABILITY:** Deidentified participant data are available from the corresponding
322 author (gkenny@uottawa.ca) upon reasonable request and signed access agreement.

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431 **TABLES****Table 1.** Study participant characteristics.

	Young Adults (n=20)	Older Adults		
		Total (n=39)	Without T2D / HTN (n=19)	With T2D / HTN (n=20)
Age, years	24.2 (3.7)	70.5 (4.3)	71.6 (4.2)	69.5 (4.2)
Sex, (F / M)	10 / 10	11 / 28	7 / 12	4 / 16
Height, cm	168.6 (8.4)	170.6 (9.4)	169.5 (10.6)	171.7 (8.3)
Body mass, kg	69.7 (14.3)	75.6 (11.9)	72.2 (11.4)	78.9 (11.8)
Body mass index, kg/m ²	24.3 (3.3)	25.9 (3.2)	25.1 (3.3)	26.7 (2.9)

Data are presented as mean (standard deviation) except for sex presented as total number. Body mass index was calculated using the following equation: weight in kilograms divided by the square of the height in meters.

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Table 2. Heart rate and heart rate variability responses in young and older adults.

Heart rate and HRV Indices	Raw data, mean (SD)		Unadjusted diff.		Adjusted diff.	
	Young (n=20)	Older (n=39)	Mean [95% CI]	P-value	Mean [95% CI]	P-value
Heart rate, beats/min						
Hour 0 (baseline)	70 (7)	66 (9)	-5 [-11, 2]	0.183	-	-
Hour 3	84 (10)	81 (14)	-3 [-10, 4]	0.398	3 [-2, 7]	0.285
Hour 6	87 (12)	83 (14)	-4 [-11, 2]	0.199	1 [-4, 6]	0.649
Hour 9	86 (12)	83 (15)	-2 [-9, 5]	0.521	3 [-2, 8]	0.177
Heart rate, % of age-predicted maximum						
Hour 0 (baseline)	36 (3)	44 (6)	8 [4, 12]	<0.001	-	-
Hour 3	43 (5)	54 (9)	11 [7, 16]	<0.001	2 [-2, 5]	0.325
Hour 6	45 (6)	55 (9)	11 [7, 15]	<0.001	1 [-2, 5]	0.475
Hour 9	44 (6)	56 (10)	12 [8, 16]	<0.001	3 [-1, 6]	0.172
Heart rate, % of reserve						
Hour 0 (baseline)	0 (0)	0 (0)	-	-	-	-
Hour 3	11 (4)	19 (12)	8 [3, 13]	0.003	-	-
Hour 6	14 (6)	21 (12)	7 [1, 13]	0.015	-	-
Hour 9	12 (6)	21 (14)	9 [4, 15]	0.002	-	-
Log SDNN						
Hour 0 (baseline)	4.2 (0.3)	3.5 (0.4)	-0.7 [-0.9, -0.5]	<0.001	-	-
Hour 3	4.1 (0.2)	3.5 (0.4)	-0.6 [-0.8, -0.4]	<0.001	-0.1 [-0.3, 0.1]	0.330
Hour 6	4.0 (0.2)	3.4 (0.4)	-0.6 [-0.8, -0.4]	<0.001	-0.1 [-0.4, 0.1]	0.206
Hour 9	4.0 (0.3)	3.4 (0.4)	-0.5 [-0.7, -0.3]	<0.001	-0.1 [-0.3, 0.1]	0.496
Log RMSSD						
Hour 0 (baseline)	3.9 (0.4)	3.1 (0.5)	-0.7 [-1.0, -0.5]	<0.001	-	-
Hour 3	3.4 (0.4)	2.8 (0.6)	-0.6 [-0.8, -0.3]	<0.001	0.1 [-0.2, 0.3]	0.452
Hour 6	3.3 (0.4)	2.7 (0.6)	-0.5 [-0.8, -0.3]	<0.001	0.1 [-0.1, 0.4]	0.345
Hour 9	3.3 (0.4)	2.7 (0.6)	-0.5 [-0.8, -0.3]	<0.001	0.1 [-0.1, 0.4]	0.328
Log HF Power						
Hour 0 (baseline)	-2.2 (0.5)	-2.5 (0.5)	-0.3 [-0.6, -0.0]	0.024	-	-
Hour 3	-2.7 (0.6)	-3.1 (0.5)	-0.3 [-0.6, -0.1]	0.017	-0.1 [-0.4, 0.1]	0.313
Hour 6	-2.6 (0.6)	-3.0 (0.6)	-0.4 [-0.7, -0.1]	0.007	-0.2 [-0.4, 0.1]	0.171
Hour 9	-2.7 (0.6)	-3.1 (0.5)	-0.4 [-0.7, -0.1]	0.003	-0.2 [-0.4, 0.0]	0.100
Cardiac vagal index						
Hour 0 (baseline)	-1.3 (0.3)	-2.0 (0.4)	-0.7 [-0.8, -0.5]	<0.001	-	-
Hour 3	-1.6 (0.2)	-2.1 (0.4)	-0.5 [-0.7, -0.3]	<0.001	0.0 [-0.2, 0.2]	0.930
Hour 6	-1.7 (0.3)	-2.2 (0.4)	-0.5 [-0.7, -0.3]	<0.001	0.0 [-0.2, 0.2]	0.917
Hour 9	-1.7 (0.3)	-2.2 (0.4)	-0.5 [-0.7, -0.3]	<0.001	0.0 [-0.2, 0.2]	0.808
DFA $\alpha 1$						
Hour 0 (baseline)	1.0 (0.2)	1.1 (0.2)	0.0 [-0.1, 0.2]	0.616	-	-
Hour 3	1.3 (0.2)	1.2 (0.3)	-0.1 [-0.2, 0.0]	0.086	-0.1 [-0.2, 0.0]	0.008
Hour 6	1.3 (0.2)	1.1 (0.3)	-0.1 [-0.3, 0.0]	0.058	-0.2 [-0.3, 0.0]	0.004
Hour 9	1.3 (0.2)	1.2 (0.3)	-0.1 [-0.3, 0.0]	0.058	-0.2 [-0.3, 0.0]	0.004

Heart rate and heart rate variability (HRV) data are presented as mean (standard deviation) of raw data and mean [95% confidence interval] of the difference in raw data between age groups (older - young). Mean differences between age groups were compared using a linear mixed-effects models both unadjusted and adjusted for baseline values. In the unadjusted models, fixed variables included age group (young and older), time (rates: hour 0, 3, 6, 9) and their interaction. In the adjusted models, baseline values were included as a continuous covariate. Standard deviation of normal RR intervals (SDNN); square root of the mean of squared differences between successive RR intervals (RMSSD); high (0.15 – 0.40 Hz) frequency band (HF); detrended fluctuation analysis used to calculate short-term fractal relationship properties of RR intervals (DFA $\alpha 1$).

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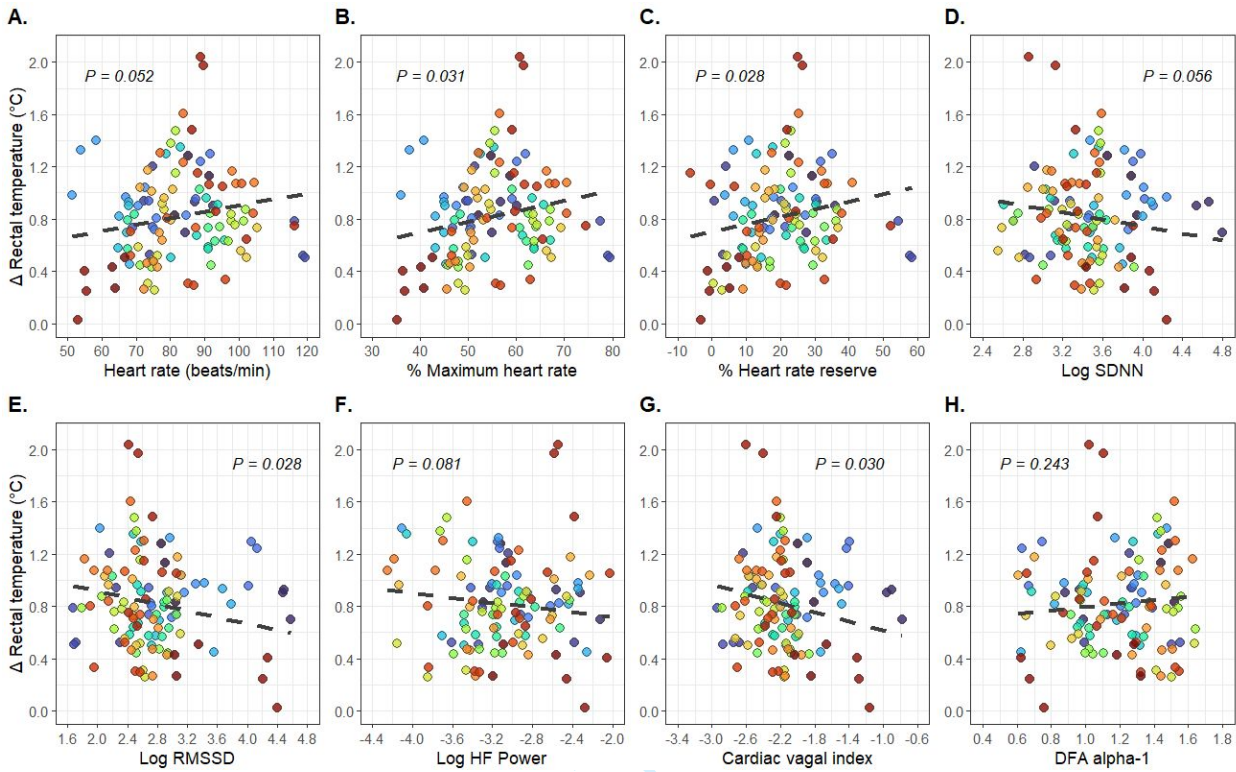
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FIGURE CAPTIONS

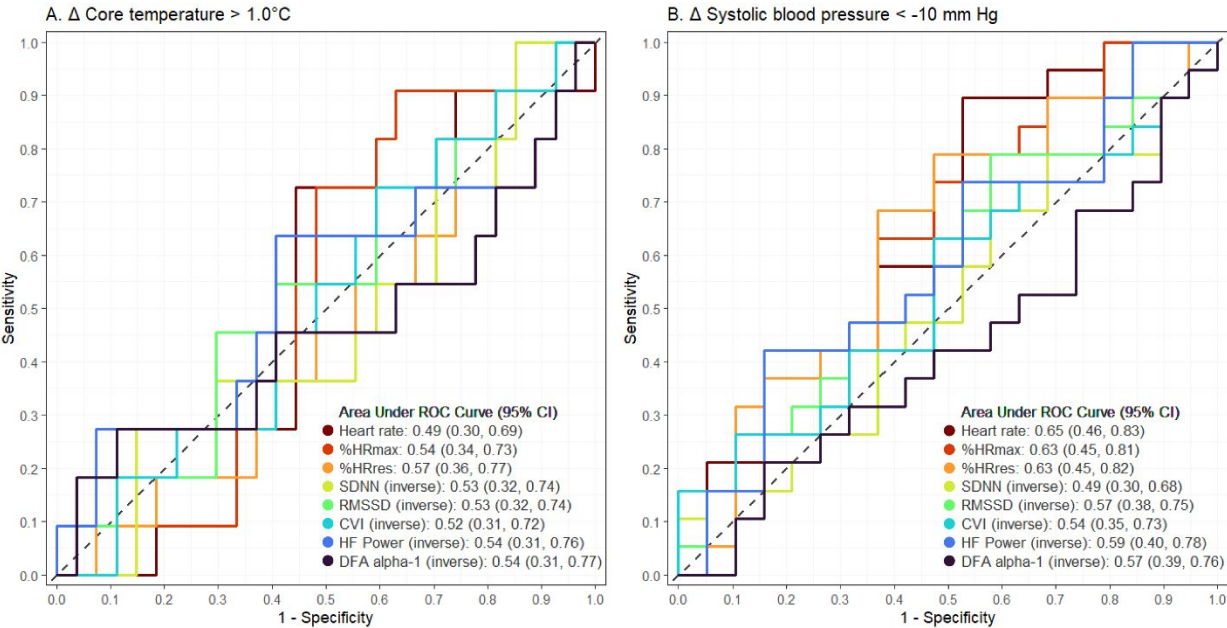
Figure 1. To examine the potential discriminatory utility of heart rate and heart rate variability (HRV) indices in differentiating elevated heat-induced physiological strain (i.e., thermal strain) in older adults, associations between each index and changes in core temperature from baseline to 3-, 6-, and 9-hours during heat exposure (40°C, 9% relative humidity) were evaluated. Heart rate and HRV indices included A) absolute heart rate, B) percentage of age-predicted maximal heart rate, C) percentage of heart rate reserve, D) log of the standard deviation of normal RR intervals (SDNN), E) log of the square root of the mean of squared differences between successive RR intervals (RMSSD), F) log of high frequency (0.15 – 0.40 Hz) power (HF), G) cardiac vagal index (CVI), and H) detrended fluctuation analysis alpha-1 (DFA α_1).

Figure 2. Receiver operating characteristics (ROC) were constructed to assess the utility of heart rate (absolute values, % age-predicted maximum heart rate, % heart rate reserve) and HRV indices measured during the final 3 hours of heat exposure (average of time points from hour 6 to 9) for identifying elevated heat-induced physiological strain (i.e., thermal strain) in older adults. The presence of heat-induced physiological strain was characterized by A) an increase in core temperature of $>1^{\circ}\text{C}$, or B) a reduction in systolic blood pressure of >10 mm Hg. Indices of HRV included the log of the standard deviation of normal RR intervals (SDNN), log of the square root of the mean of squared differences between successive RR intervals (RMSSD), log of high frequency (0.15 – 0.40 Hz) power (HF), cardiac vagal index (CVI), and detrended fluctuation analysis alpha-1 (DFA α_1).

Figure 1.



461 **Figure 2.**



462

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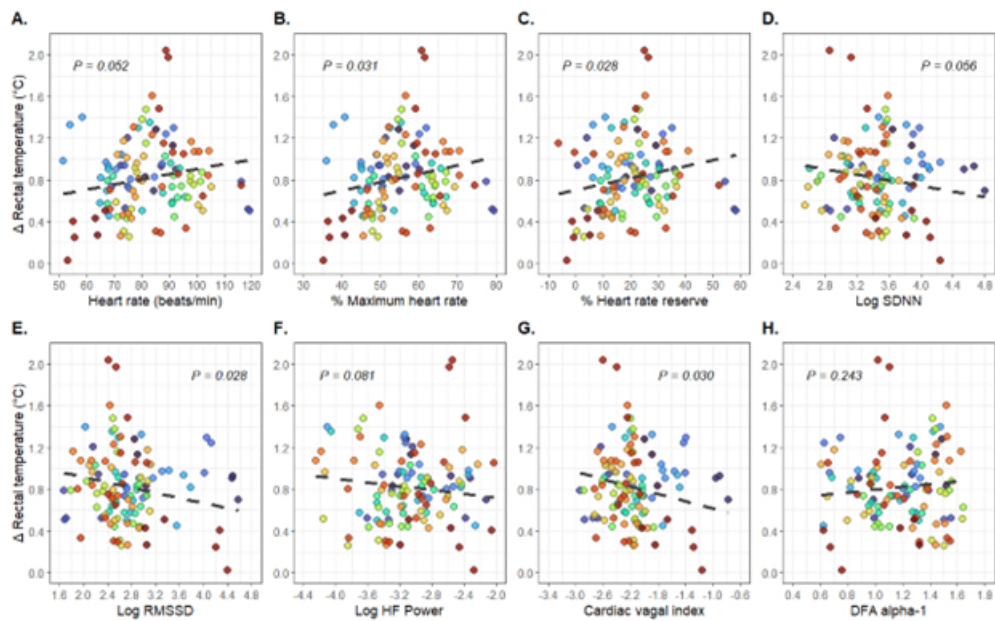


Figure 1.

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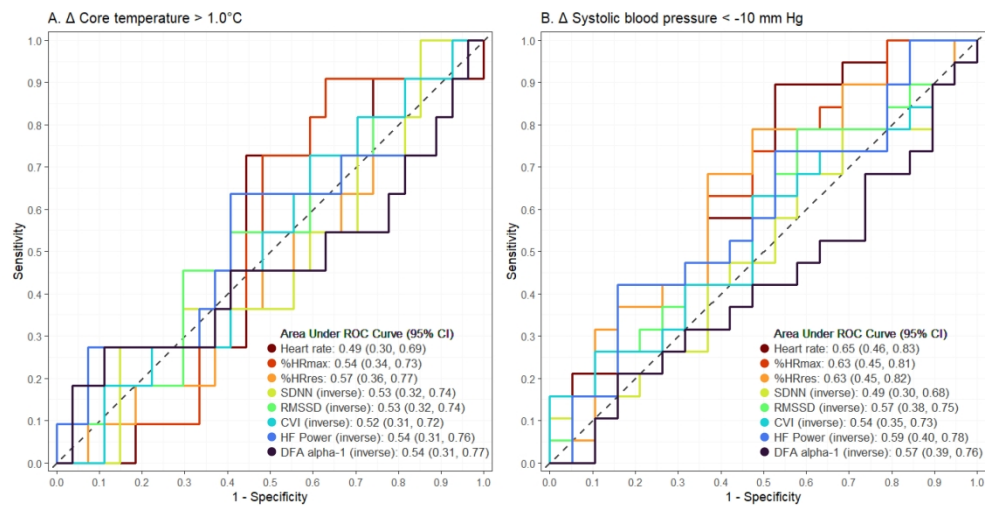


Figure 2

758x393mm (118 x 118 DPI)